

(RAPTOR)

EXPERIMENT 4:

The string is a sequence of characters placed in double quotes (" ").
Performing different operations on string data is called String Handling. Strings are immutable. Whenever a change to a String is made, an entirely new String is created. If we want to store a group of characters we can use a char array. String provides various methods to perform different operations on strings. Using Raptor draw a flowchart to display the total number of characters in the string and returns it.

Aim

The objective of this experiment is to create a flowchart using **RAPTOR** to accept a string as input, calculate the total number of characters in the string, and display the result.

Software Requirements

Software: RAPTOR Flowchart Software

Operating System: Windows 10/11

Hardware: PC/Laptop

Input Device: Keyboard

Theory

RAPTOR is a graphical programming tool used to design algorithms using flowcharts. It allows users to develop logic without writing programming code. In this experiment, a string is accepted as input, and the built-in **LengthOf()** function is used to determine the total number of characters present in the string. The result is then displayed on the screen. This experiment helps understand string manipulation and the use of predefined functions in RAPTOR.

Procedure

Step 1:

Open the **RAPTOR** application and create a new flowchart.

Step 2:

Insert an **Input** symbol and prompt the user to enter a string.

Step 3:

Insert an **Assignment** symbol and use the **LengthOf()** function to calculate the total number of characters in the entered string.

Step 4:

Insert an **Output** symbol to display the message "The total number of characters is" along with the calculated length.

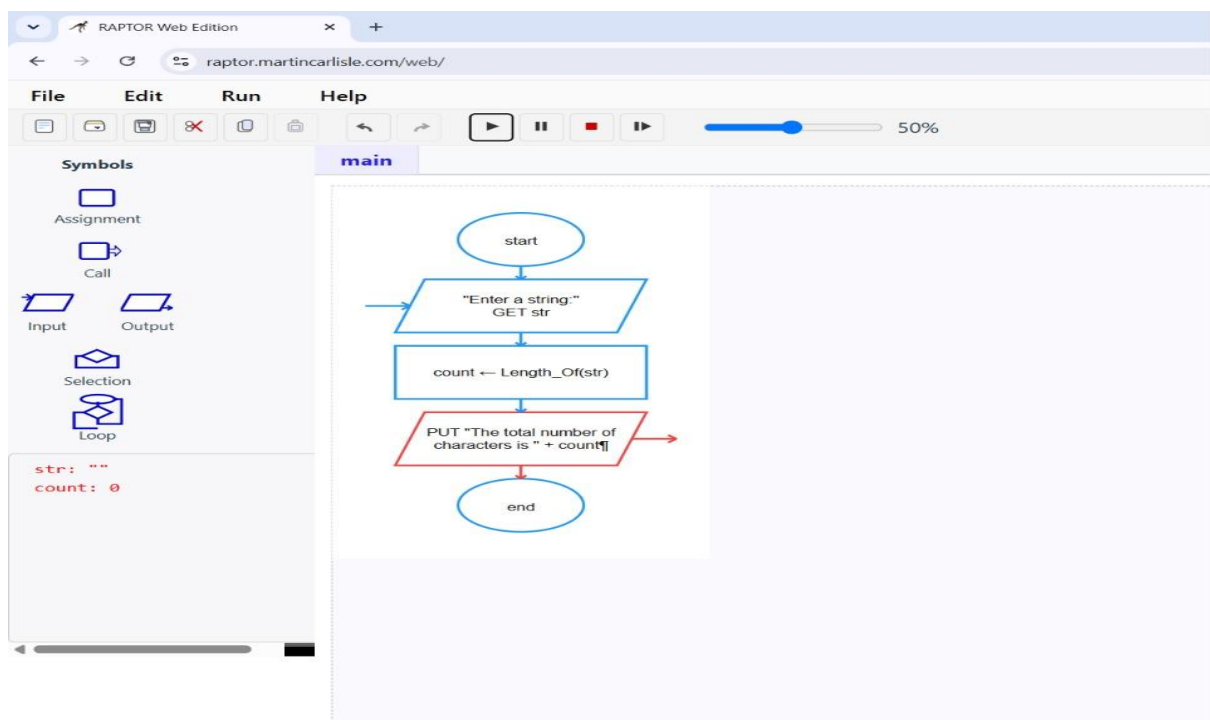


Figure 1. Displaying the Result.

Step 5:

Run the flowchart, enter different strings, and verify that the output displays the correct number of characters.

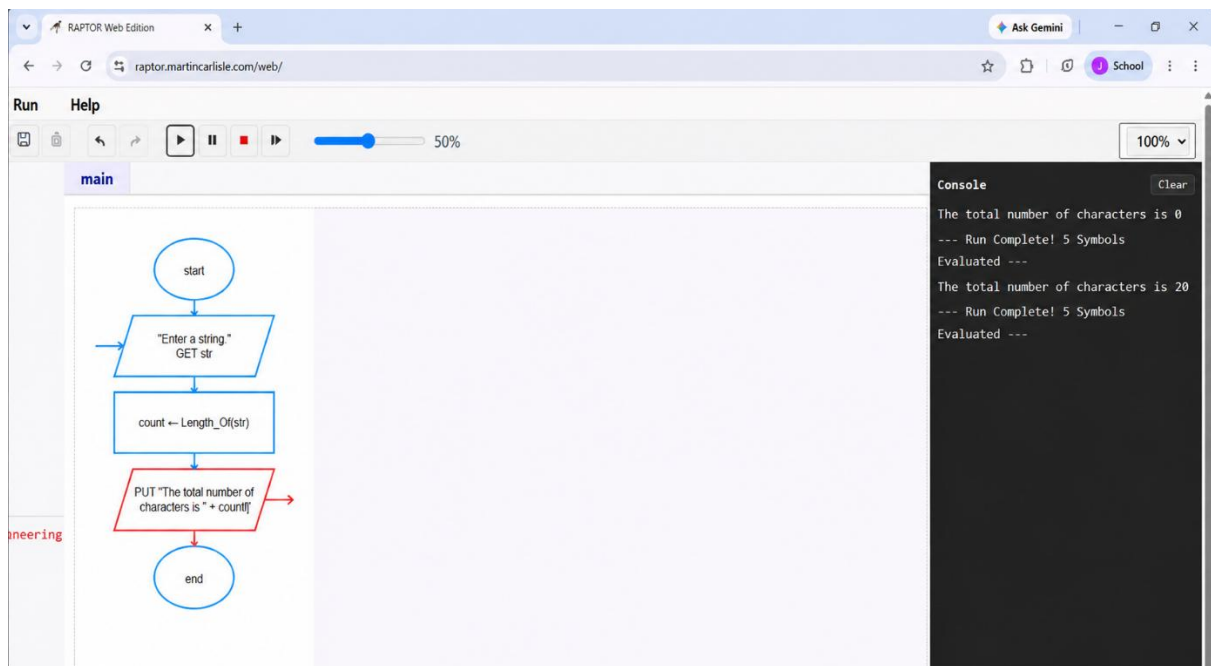


Figure 2. Executing the Flowchart.

Result

The flowchart was successfully created using **RAPTOR** to calculate the total number of characters in a given string. The program correctly accepted the input string, computed its length using the **LengthOf()** function, and displayed the result. The experiment demonstrated the use of string functions and flowchart-based programming for solving simple string-processing problems.